

CS714: Secure Computation

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Definition

A probability space is a triple $(S, \mathcal{F}, \text{Pr})$ where:

- S is the **sample space**
- $\mathcal{F}$ is the **event space**
- Pr is a probability measure

Probability Space

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Example

- $S = \{1, 2, 3, 4, 5, 6\}$ (fair die)
- $\mathcal{F} = 2^S$
- $\text{Pr}[\{i\}] = \frac{1}{6}$

Key Fact

If S is countable, then for any event $E \subseteq S$:

$$\Pr[E] = \sum_{e \in E} \Pr[e]$$

Example

Let $E = \{2, 4, 6\}$.

$$\Pr[E] = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

Random Variables

Definition

A random variable is a function:

$$X : S \rightarrow A$$

where A is called the **range**.

Example

Let:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Define:

$$X(s) = s \bmod 2$$

- $X = 0$ for even outcomes
- $X = 1$ for odd outcomes

Events of the Form $X = x$

Definition

The event $X = x$ means:

$$X^{-1}(x) = \{s \in S \mid X(s) = x\}$$

Example

What is $X^{-1}(0)$? (Here X^{-1} is pre-image!)

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Example

$$X^{-1}(0) = \{2, 4, 6\}$$

$$\Pr[X = 0] = \Pr[\{2, 4, 6\}] = \frac{1}{2}$$

Events of the Form $X \in C$

Definition

For $C \subseteq A$: $X \in C := \bigcup_{x \in C} X^{-1}(x)$

Example

Let: $C = \{0\}$

Example

$$X \in C = \{2, 4, 6\}$$

$$\Pr[X \in C] = \Pr[X = 0] = \frac{1}{2}$$

Support of a Random Variable

Definition

The support of X is:

$$\text{supp}(X) = \{x \mid \Pr[X = x] > 0\}$$

Example

For the die parity variable:

$$\text{supp}(X) = \{0, 1\}$$

Predicate Random Variables

Definition

A predicate random variable has range $\{\perp, \top\}$.

We write:

$$\Pr[P] := \Pr[P = \top]$$

What is $\Pr[P]$?

$$\text{Let: } P(s) = \begin{cases} \top & \text{if } s \in \{1, 2\} \\ \perp & \text{otherwise} \end{cases}$$

Example

$$\Pr[P] = \frac{2}{6} = \frac{1}{3}$$

Combining Random Variables

Definition

If:

$$X : S \rightarrow A, \quad Y : S \rightarrow B$$

and

$$g : A \times B \rightarrow C$$

then:

$$g(X, Y)(s) = g(X(s), Y(s))$$

Example

Two dice:

$$Z = X + Y$$

where Z is the sum.

Combining Random Variables

Basic Idea

- Given random variables $X : S \rightarrow A$, $Y : S \rightarrow B$
- Define a new random variable $Z = g(X, Y)$ for any deterministic function $g : A \times B \rightarrow C$.

Interpretation

Combining random variables: *applying a function pointwise on outcomes in the sample space.*

Combining via Arithmetic Operations

Example: Sum

Let

$$Z = X + Y$$

where $X, Y : S \rightarrow \mathbb{Z}_p$. Then

$$Z(s) = X(s) + Y(s) \pmod{p}.$$

Distribution

$$\Pr[Z = z] = \sum_{x+y=z} \Pr[X = x, Y = y]$$

Joint Random Variables

Joint Variable

The pair (X, Y) is itself a random variable:

$$(X, Y) : S \rightarrow A \times B$$

defined by

$$(X, Y)(s) = (X(s), Y(s)).$$

Joint Distribution

$$\Pr[X = x, Y = y] = \Pr(X^{-1}(x) \cap Y^{-1}(y))$$

Example: Joint Random Variables

Question

Two fair coins are tossed.

Let

X = outcome of the first toss, Y = outcome of the second toss.

What is the joint distribution of (X, Y) ?

Answer

The joint random variable (X, Y) takes values in $\{H, T\} \times \{H, T\}$.

$$\Pr[X = H, Y = H] = \Pr[X = H, Y = T] = \Pr[X = T, Y = H] = \Pr[X = T, Y = T] = \frac{1}{4}.$$

Assignment Notation

Important

The notation:

$$Y \leftarrow Z$$

means **definition**, not randomness.

Example

If:

$$Y \leftarrow 2 \cdot X^2$$

then:

$$\Pr[X^2 = Y/2] = 1$$

Statistical Distance

Definition

Let X_0 and X_1 be random variables with the same range D . The **statistical distance** between X_0 and X_1 is:

$$\Delta(X_0, X_1) = \frac{1}{2} \sum_{d \in D} |\Pr[X_0 = d] - \Pr[X_1 = d]|$$

Upper Bound

What is the upper bound of $\Delta(X_0, X_1)$?

Statistical Distance: Upper Bound

For any $d \in D$,

$$|\Pr[X_0 = d] - \Pr[X_1 = d]| \leq \Pr[X_0 = d] + \Pr[X_1 = d].$$

Summing over $d \in D$:

$$\sum_{d \in D} |\Pr[X_0 = d] - \Pr[X_1 = d]| \leq 2.$$

Hence:

$$0 \leq \Delta(X_0, X_1) \leq 1.$$

When Is Distance Equal to 1?

$$\Delta(X_0, X_1) = 1 \iff \text{Supports of } X_0 \text{ and } X_1 \text{ are disjoint.}$$

That is:

$$\forall d \in D, \Pr[X_0 = d] \cdot \Pr[X_1 = d] = 0.$$

One sample distinguishes with certainty.

Example: Three-Face Die

Let $D = \{1, 2, 3\}$.

Define:

$$\Pr[X_0 = 1] = 1, \quad \Pr[X_1 = 3] = 1.$$

Then:

$$\Delta(X_0, X_1) = \frac{1}{2}(|1 - 0| + |0 - 0| + |0 - 1|) = 1.$$

Maximum possible statistical distance.

Intuition Behind Statistical Distance

Intuition

Statistical distance measures how well one can distinguish two distributions by observing a single sample.

- $\Delta = 0 \Rightarrow$ distributions are identical
- $\Delta = 1 \Rightarrow$ distributions are completely disjoint

Cryptographic Meaning

Maximum distinguishing advantage of any adversary is exactly $\Delta(X_0, X_1)$.

Example 1: Identical Distributions

Setup

Let:

$$X_0 \sim \text{Uniform}(\{0, 1\}), \quad X_1 \sim \text{Uniform}(\{0, 1\})$$

Probabilities

$$\Pr[X_0 = 0] = \Pr[X_1 = 0] = \frac{1}{2}$$

$$\Pr[X_0 = 1] = \Pr[X_1 = 1] = \frac{1}{2}$$

Distance

$$\Delta(X_0, X_1) = 0$$

Example 2: Slightly Different Distributions

Setup

Let $D = \{0, 1\}$ and:

$$\Pr[X_0 = 0] = \frac{1}{2}, \quad \Pr[X_0 = 1] = \frac{1}{2}$$

$$\Pr[X_1 = 0] = \frac{3}{4}, \quad \Pr[X_1 = 1] = \frac{1}{4}$$

Distance Computation

$$\Delta(X_0, X_1) = \frac{1}{2} \left(\left| \frac{1}{2} - \frac{3}{4} \right| + \left| \frac{1}{2} - \frac{1}{4} \right| \right) = \frac{1}{4}$$

Example 3: Disjoint Distributions

Setup

Let:

$$X_0 = 0 \text{ always, } X_1 = 1 \text{ always}$$

Probabilities

$$\Pr[X_0 = 0] = 1, \quad \Pr[X_1 = 1] = 1$$

Distance

$$\Delta(X_0, X_1) = 1$$

Interpretation

Perfect distinguishability.

Equivalent Definition (Using Events)

Theorem

$$\Delta(X_0, X_1) = \max_{C \subseteq D} |\Pr[X_0 \in C] - \Pr[X_1 \in C]|$$

Meaning

The statistical distance equals the maximum probability gap over all events of the form $X \in C$.

Example Using $X \in C$

Setup

From earlier example:

$$\Pr[X_0 = 0] = \frac{1}{2}, \quad \Pr[X_1 = 0] = \frac{3}{4}$$

Choice of Event

Choose:

$$C = \{0\}$$

Distinguishing Advantage

$$|\Pr[X_0 \in C] - \Pr[X_1 \in C]| = \left| \frac{1}{2} - \frac{3}{4} \right| = \frac{1}{4}$$

Statistical Distance and Predicates

Key Connection

For any predicate P :

$$|\Pr[P(X_0)] - \Pr[P(X_1)]| \leq \Delta(X_0, X_1)$$

Cryptography View

No test (predicate) can distinguish X_0 and X_1 with advantage greater than their stat distance.

One-Time Pad (OTP)

Setting

- Message space: $\{0, 1\}^n$
- Key space: $\{0, 1\}^n$
- Key $K \leftarrow U_n$ (used once)

Encryption

$$\text{Enc}_K(M) = M \oplus K$$

Decryption

$$\text{Dec}_K(C) = C \oplus K$$

The adversary observes:

$$C = M \oplus K$$

where:

- M is unknown
- K is uniformly random

Question

Does C reveal anything about M ?

Induced Ciphertext Distribution

Fix any message $m \in \{0, 1\}^n$.

Let:

$$C_m := m \oplus U_n$$

Since XOR with a fixed string is a permutation:

$$C_m \sim U_n$$

Thus, for every message m :

$$\Pr[C_m = c] = 2^{-n} \quad \forall c$$

Two Messages, Same Distribution

Let $m_0, m_1 \in \{0, 1\}^n$ be any two messages.

Then:

$$C_{m_0} = m_0 \oplus U_n \quad C_{m_1} = m_1 \oplus U_n$$

Both are uniformly distributed over $\{0, 1\}^n$.

For any (even unbounded) distinguisher $\mathcal{D}$:

$$\Pr[\mathcal{D}(C_{m_0}) = 1] = \Pr[\mathcal{D}(C_{m_1}) = 1]$$

Therefore:

$$\text{Adv}_{\mathcal{D}}(C_{m_0}, C_{m_1}) = 0$$

The statistical distance is:

$$\Delta(C_{m_0}, C_{m_1}) = 0$$

Perfect Secrecy

- Ciphertexts for different messages are **identically distributed**
- The adversary gains **zero information**

Probabilistic Model

- Cryptographic algorithms are inherently randomized
 - Security is defined over *distributions*, not single runs
 - We need a formal model to reason about randomness and outputs
-
- Algorithm A is a deterministic TM with an extra random tape
 - Random tape: $r \in \{0, 1\}^\infty$
 - Inputs and randomness are separated

Algorithm Execution

For inputs $x_1, \dots, x_\ell$ and randomness r : $A(x_1, \dots, x_\ell; r)$ denotes the output of A on random tape r .

Fix inputs $x_1, \dots, x_\ell$.

- Sample $r \leftarrow \{0, 1\}^\infty$
- $A(x_1, \dots, x_\ell)$ is the induced random variable

For any output y : $\Pr[A(x_1, \dots, x_\ell) = y] = \Pr_r[A(x_1, \dots, x_\ell; r) = y]$

Termination Convention

- Algorithms terminate with probability 1
- Non-termination is mapped to a special symbol $\perp$
- Ensures well-defined random variables

Distinguishing Algorithm

A **distinguisher** is a probabilistic algorithm

$$\mathcal{D} : \mathcal{X} \rightarrow \{0, 1\}$$

Interpretation:

- $\mathcal{D}(x) = 1$ means “guess X_1 ”
- $\mathcal{D}(x) = 0$ means “guess X_0 ”

Distinguishing Algorithm

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- $\mathcal{D}(x) = 1$ means “guess X_1 ”
- $\mathcal{D}(x) = 0$ means “guess X_0 ”

$\mathcal{D}$ succeeds if its output depends on which distribution generated x .

Distinguishing Advantage

The **distinguishing advantage** of $\mathcal{D}$ is:

$$\text{Adv}_{\mathcal{D}}(X_0, X_1) = |\Pr[\mathcal{D}(X_0) = 1] - \Pr[\mathcal{D}(X_1) = 1]|$$

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Interpretation:

- Advantage = 0 : perfect indistinguishability
- Larger advantage : easier to tell apart

Why This Measure?

If $\mathcal{D}$ cannot distinguish:

$$\Pr[\mathcal{D}(X_0) = 1] \approx \Pr[\mathcal{D}(X_1) = 1]$$

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If $\mathcal{D}$ cannot distinguish:

$$\Pr[\mathcal{D}(X_0) = 1] \approx \Pr[\mathcal{D}(X_1) = 1]$$

The absolute difference:

- ignores labeling conventions,
- measures *maximum bias*,
- matches hypothesis testing intuition.

Baseline: Random Guessing

If $\mathcal{D}$ ignores its input and guesses randomly:

$$\Pr[\mathcal{D}(X_0) = 1] = \Pr[\mathcal{D}(X_1) = 1] = \frac{1}{2}$$

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If $\mathcal{D}$ ignores its input and guesses randomly:

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Thus:

$$\text{Adv}_{\mathcal{D}} = 0$$

Security means:

No efficient algorithm does *significantly better*.

Security via Indistinguishability

We say X_0 and X_1 are **computationally indistinguishable** if:

$$\forall \text{ PPT } \mathcal{D}, \quad \text{Adv}_{\mathcal{D}}(X_0, X_1) \leq \text{negl}(\lambda)$$

Security via Indistinguishability

We say X_0 and X_1 are **computationally indistinguishable** if:

$$\forall \text{ PPT } \mathcal{D}, \quad \text{Adv}_{\mathcal{D}}(X_0, X_1) \leq \text{negl}(\lambda)$$

Meaning:

- no efficient algorithm can reliably tell them apart

How Security Proofs Work

Typical proof structure:

- 1 Assume a distinguisher $\mathcal{D}$ with non-negligible advantage
- 2 Use $\mathcal{D}$ as a subroutine
- 3 Construct an algorithm that breaks a known assumption

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Conclusion:

Such a distinguisher cannot exist.

Why Do We Need a Formal Security Definition?

Informal statements such as:

- “The scheme looks random”
- “The adversary learns nothing”

are not mathematically precise.

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are not mathematically precise.

Goal: Define security in a way that

- can be stated precisely,
- can be proven rigorously,
- rules out *all efficient attacks*.

Security as a Guessing Game

We model an attacker as an algorithm that:

- sees some data,
- tries to guess how that data was generated.

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Security means:

No efficient algorithm can guess significantly better than chance.

IND-CPA = Distinguishability

- Adversary chooses (m_0, m_1)
- Challenger encrypts m_b
- Adversary guesses b

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Advantage

$$\text{Adv}^{\text{IND-CPA}} = \left| \Pr[b' = b] - \frac{1}{2} \right|$$

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Advantage

$$\text{Adv}^{\text{IND-CPA}} = \left| \Pr[b' = b] - \frac{1}{2} \right|$$

Security Meaning

Encryption is IND-CPA secure iff no PPT distinguisher has non-negligible advantage.

Indistinguishability-Based Encryption Security

Let $(\text{KeyGen}, \text{Enc}, \text{Dec})$ be a public-key encryption scheme.

Security is defined via a *game* between:

- a challenger, and
- an adversary trying to distinguish encryptions.

We compare two threat models:

IND-CPA vs. **IND-CCA.**

IND-CPA Game

- 1 Challenger runs $(pk, sk) \leftarrow \text{KeyGen}$ and gives pk to $\mathcal{A}$.
- 2 $\mathcal{A}$ may query an *encryption oracle* $\text{Enc}(pk, \cdot)$.
- 3 $\mathcal{A}$ submits two messages (m_0, m_1) of equal length.
- 4 Challenger samples $b \leftarrow \{0, 1\}$ and gives

$$c^* = \text{Enc}(pk, m_b).$$

- 5 $\mathcal{A}$ outputs a guess b' .

$$\text{Adv}_{\mathcal{A}}^{\text{IND-CPA}} = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

IND-CCA Game

Same as IND-CPA, except:

- $\mathcal{A}$ also has access to a *decryption oracle*

$\text{Dec}(sk, \cdot),$

- with the restriction that it cannot query $\text{Dec}(sk, c^*)$.

$$\text{Adv}_{\mathcal{A}}^{\text{IND-CCA}} = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

What IND-CCA Captures

Intuition

IND-CCA models *active attackers* who can tamper with ciphertexts and observe how they decrypt.

- Strong notion for real-world security
- Protects against malleability and padding attacks
- Required for secure protocols and APIs

Example: Encrypt-then-MAC, CCA-secure KEMs.

Relationship and Takeaway

$$\text{IND-CCA} \implies \text{IND-CPA}$$

- IND-CPA is necessary but not sufficient
- IND-CCA is the gold standard for encryption

Key Takeaway

IND-CPA hides *plaintexts*. IND-CCA hides plaintexts *even under active attacks*.

Two Competing Distributions

Let:

- X_0 and X_1 be random variables
- Both range over the same domain $\mathcal{X}$

Examples:

- output of a cryptographic algorithm
- output of a truly random process

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Examples:

- output of a cryptographic algorithm
- output of a truly random process

Question:

Can an algorithm tell which one it is seeing?

Key Takeaways

- Security is defined via **distinguishing experiments**
- Advantage measures how well an attack works
- “Secure” means *negligible advantage for all efficient algorithms*

Pseudorandom Generators

A function

$$G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}, \quad \ell(n) > n$$

is a *pseudorandom generator (PRG)* if its output is computationally indistinguishable from uniform.

Real vs Random Experiments

Real World

$s \leftarrow \{0, 1\}^n$, output $G(s)$ (Pseudorandom)

Ideal World

$r \leftarrow \{0, 1\}^{\ell(n)}$, output r (True Randomness)

$\text{Real}_G(n)$ vs $\text{Rand}(n)$

Distinguishing Algorithm for PRGs

Recall

- A distinguisher D is a PPT algorithm
- Input: a string $y \in \{0, 1\}^{\ell(n)}$
- Output: a bit in $\{0, 1\}$

$$\text{Adv}_G^{\text{prg}}(D, n) = |\Pr[D(G(s)) = 1] - \Pr[D(r) = 1]|$$

where:

- $s \leftarrow \{0, 1\}^n$
- $r \leftarrow \{0, 1\}^{\ell(n)}$

PRG Security Definition

Definition

G is a PRG if for every PPT distinguisher D ,

$$\text{Adv}_G^{\text{prg}}(D, n) \leq \text{negl}(n)$$

Interpretation

- Distinguisher formalizes *any feasible statistical test*
- If no PPT D can distinguish, $G(s)$ “looks random”
- This is computational indistinguishability

Computational vs. Statistical Indistinguishability

Give example of two families of random variables that are perfectly distinguishable statistically, however are computationally indistinguishable.

Distinguishing Advantage

Let $\mathcal{A}$ be a (possibly randomized) algorithm and let

$$X_0 = \{X_0(\kappa)\}_{\kappa \in \mathbb{N}}, \quad X_1 = \{X_1(\kappa)\}_{\kappa \in \mathbb{N}}$$

be families of random variables.

Definition of Advantage

For each security parameter κ ,

$$\text{Adv}_{\mathcal{A}}(X_0, X_1)(\kappa) \stackrel{\text{def}}{=} \text{Adv}_{\mathcal{A}(\kappa, \cdot)}(X_0(\kappa), X_1(\kappa)),$$

where $\mathcal{A}(\kappa, \cdot)$ denotes $\mathcal{A}$ with κ fixed and one sample as input.

Distinguishing Advantage of a Class of Algorithms

In cryptography, we are typically not interested if *one particular algorithm* can distinguish a class of random variables.

Indistinguishability by a Class of Algorithms

Let $\mathcal{A}$ be a class of distinguishers.

We say X_0 and X_1 are *indistinguishable by $\mathcal{A}$* if

$$\mathcal{A}(X_0) \equiv^{\text{stat}} \mathcal{A}(X_1) \quad \text{for all } \mathcal{A} \in \mathcal{A}.$$

$$X_0 \equiv^{\mathcal{A}} X_1$$

Why OTP Is Stronger Than PRGs

- OTP achieves **statistical** (perfect) indistinguishability
- Security holds against **all** adversaries
- No computational assumptions required

In contrast:

- PRGs achieve only computational indistinguishability

Indistinguishability: Unified View

Two distributions D_0, D_1 over the same domain.

A distinguisher A outputs 1 if it believes the sample came from D_0 .

$$\text{Adv}_A(D_0, D_1) = |\Pr[A(D_0) = 1] - \Pr[A(D_1) = 1]|$$

Perfect indistinguishability:

$$\Pr[D_0 = x] = \Pr[D_1 = x] \quad \forall x$$

$\Rightarrow \text{Adv}_A = 0$ for all (unbounded) A

Example: One-Time Pad

Ciphertext distributions are identical.

Statistical indistinguishability:

$$\Delta(D_0, D_1) \leq \text{negl}(n)$$

Allows tiny, information-theoretically negligible leakage.

Statistical Example: Truncated PRG

Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a PRG.

- D_0 : first n bits of $G(s)$, $s \leftarrow \{0, 1\}^n$
- D_1 : uniform over $\{0, 1\}^n$

For each $y \in \{0, 1\}^n$:

$$\Pr[D_0 = y] = 2^{-n} \pm \text{negl}(n)$$

$\Rightarrow$ Statistical but not perfect

Why Full PRG Is Not Statistical

$$|\text{Supp}(D_0)| \leq 2^n, \quad |\text{Supp}(D_1)| = 2^{2n}$$

For $x \notin \text{Im}(G)$:

$$\Pr[D_0 = x] = 0, \quad \Pr[D_1 = x] = 2^{-2n}$$

$$\Delta(D_0, D_1) \geq \frac{1}{2}(1 - 2^{-n}) \approx 1$$

PRGs are only computationally secure.

Hierarchy and Takeaway

Perfect $\subsetneq$ Statistical $\subsetneq$ Computational

- Perfect: distributions identical
- Statistical: negligible information leakage
- Computational: leakage exists but is efficiently hidden

Interactive Agents and Ports

Interactive Agent

- An *interactive agent* has:
 - a set of **input ports** $\text{In}(A)$
 - a set of **output ports** $\text{Out}(A)$
- Ports are identified by *names*
- Agents interact by connecting outputs to inputs with the same name

Port Compatibility

Interactive agents $A_1, \dots, A_n$ are *port compatible* if for all $i \neq j$:

$$\text{In}(A_i) \cap \text{In}(A_j) = \emptyset \quad \text{and} \quad \text{Out}(A_i) \cap \text{Out}(A_j) = \emptyset$$

- No two agents have the same input port name
- No two agents have the same output port name